



The Mass-Energy-Information Equivalence: A bottom-up identification of the particle spectrum via FCC lattice error correction

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ABSTRACT

Does information possess physical mass? Modelling the physical vacuum as a substrate-free quantum error-correcting code suggests that an elementary particle's mass is simply its fault-tolerant verification cost. We test this Mass-Energy-Information (M/E/I) equivalence on the Face-Centred Cubic (FCC) lattice, tracking defects within a $[[192, 130, 3]]$ CSS code. Through a bottom-up classification of all possible defect geometries, we filter 25 candidate states through four strict thermodynamic and topological axioms — Minimum Topological Dimension, Sector Completeness, Boundary Closure, and Kinematic Shedding — each derived from established QEC theory or lattice gauge theory. Exactly 5 physically stable states survive this sieve. Their verification costs — 1, 207, 273, 1836, and 1839 — match the empirical mass ratios of the electron, muon, pion, proton, and neutron to within 0.12%. The rejected configurations violate specific physical constraints and match no known particles. No parameters are fitted. This offers highly constrained macroscopic evidence that inertial rest mass is the thermodynamic shadow of quantum error correction overhead.

1. Introduction

Landauer's 1961 theorem [1] cemented the idea that information has a physical footprint: erasing or manipulating one bit dissipates at least $E \geq kT \ln 2$ of energy. Coupling this with Einstein's mass-energy equivalence [2] implies a Mass-Energy-Information (M/E/I) equivalence, where one bit carries a fundamental mass $m = kT \ln 2 / c^2$.

Measuring this mass in a laboratory remains practically impossible — the Landauer mass near 10^{-35} kg at room temperature is drowned out by any material substrate [3]. We can instead investigate the M/E/I equivalence in a system lacking an independent substrate: the physical vacuum itself. If spacetime is modelled as a discrete quantum error-correcting network — a CSS stabilizer code on the FCC lattice [4] — particles emerge as localized syndrome patterns. Without a hardware substrate, their informational verification cost is their only source of mass.

An interactive 3D visualization of the FCC lattice code and the five surviving defect structures is available at https://raghu91302.github.io/ssmtheory/fcc_interactive_5.html.

This paper presents: (i) the physical motivation for vacuum QEC and the mechanism of emergent Lorentz invariance (Section 2); (ii) a strict mathematical definition of C_x grounded in QEC theory (Section 3); (iii) physical derivations of all four axioms (Section 5); (iv) an enumeration of 25 candidate defect geometries (Section 6); (v) step-by-step derivations of all five surviving particles (Section 7); (vi) connections to the broader Standard Model (Section 9).

2. Physical motivation: Why does the vacuum perform QEC?

2.1. Error source and stability

The FCC lattice is the unique densest sphere packing in three dimensions [5]. A ground state of maximum packing density is also maximally stable: any deviation from perfect close packing costs energy proportional to the local density defect. We interpret this stability requirement as continuous quantum error correction. The error source is not an external thermal bath or vacuum fluctuations in the conventional QFT sense, but the intrinsic discretization of spacetime: any departure from exact FCC coordination constitutes a topological mismatch — a logical error — that costs energy $C_x \times kT \ln 2$ to detect and restore. The FCC geometry is simultaneously the ground state and the error-correcting code.

2.2. Emergent Lorentz invariance

A common concern about discrete lattice models is conflict with special relativity. We establish emergent Lorentz invariance in two steps: (i) exact spatial isotropy from the structure tensor, and (ii) emergent Lorentz boosts in the continuum limit.

Step 1: Spatial isotropy. The $K = 12$ FCC nearest-neighbour bond vectors are

$$\mathbf{n}_j \in \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\} / \sqrt{2}, \quad j = 1, \dots, 12. \quad (1)$$

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Define the rank-2 structure tensor $S_{\mu\nu} \equiv \sum_{j=1}^{12} n_j^\mu n_j^\nu$. We compute each component explicitly:

$$S_{xx} = \sum_j (n_j^x)^2 = \underbrace{4 \times \frac{1}{2}}_{\text{bonds } (\pm 1, \pm 1, 0)} + \underbrace{4 \times \frac{1}{2}}_{\text{bonds } (\pm 1, 0, \pm 1)} + \underbrace{0}_{\text{bonds } (0, \pm 1, \pm 1)} = 4,$$

$$S_{xy} = \sum_j n_j^x n_j^y = \frac{1}{2} [(+1)(+1) + (+1)(-1) + (-1)(+1) + (-1)(-1)] + 0 + 0 = 0.$$

By the three-fold permutation symmetry of the bond set, $S_{yy} = S_{zz} = 4$ and all off-diagonal components vanish. Therefore:

$$S_{\mu\nu} = 4 \delta_{\mu\nu}. \quad [\text{exact, by enumeration}] \quad (2)$$

This guarantees equal propagation speed in every spatial direction.

The odd-rank tensor $T_{\mu\nu\lambda} \equiv \sum_j n_j^\mu n_j^\nu n_j^\lambda$ vanishes exactly: every bond $\mathbf{n}_j$ has a partner $-\mathbf{n}_j$ (the FCC lattice is centrosymmetric), so every odd-power sum cancels:

$$T_{\mu\nu\lambda} = 0. \quad [\text{exact, by inversion symmetry}] \quad (3)$$

Eq. (3) ensures $\omega(\mathbf{k}) = \omega(-\mathbf{k})$: there is no preferred direction and no linear (k) term in the dispersion.

Step 2: Isotropy of the dispersion relation. For a scalar field on the FCC lattice, the dispersion relation is $\omega(\mathbf{k})^2 = \kappa \sum_{j=1}^{12} [1 - \cos(\mathbf{k} \cdot \mathbf{n}_j a)]$. At long wavelengths ($|\mathbf{k}|a \ll 1$), expand the cosine:

$$\omega(\mathbf{k})^2 \approx \frac{\kappa a^2}{2} \sum_{j=1}^{12} (\mathbf{k} \cdot \mathbf{n}_j)^2 = \frac{\kappa a^2}{2} k_\mu k_\nu S_{\mu\nu} = \frac{\kappa a^2}{2} \cdot 4 |\mathbf{k}|^2 = 2\kappa a^2 |\mathbf{k}|^2, \quad (4)$$

so $\omega = c_{\text{lat}} |\mathbf{k}|$ with $c_{\text{lat}} = a\sqrt{2\kappa}$. The dispersion is exactly isotropic at all orders in k for which the Taylor expansion is valid — the isotropy is a direct algebraic consequence of $S_{\mu\nu} = 4\delta_{\mu\nu}$, not an approximation. Corrections appear only at $O(k^4 a^4)$ from higher terms in the cosine expansion, suppressed by $(|\mathbf{k}|a)^4 \sim (E/M_P)^4$ for Planck-scale lattice spacing $a \sim \ell_P$.

Step 3: Lorentz boosts. The two tensor identities (2)–(3) and the isotropic linear dispersion (4) are sufficient for Lorentz boosts to emerge in the standard continuum limit [6]. At all energies below the lattice cutoff $1/a$ (taken to be the Planck scale $M_P \sim 10^{19}$ GeV), these corrections are negligible and the dispersion is exactly isotropic. Lorentz boosts then emerge in the standard way: a lattice Hamiltonian whose dispersion is isotropic and linear in k (at long wavelengths) gives rise to a relativistic effective field theory in the continuum limit [6]. The lattice does break Lorentz symmetry at the cutoff scale $1/a$, but these violations are suppressed by $(E/M_P)^2$ and are consistent with all current experimental bounds [7], which constrain Lorentz violation below 10^{-20} – 10^{-40} of the Planck scale.

3. Thermodynamic cost in a substrate-free vacuum

3.1. Strict definition of C_x

Definition 1 (Fault-Tolerant Verification Cost). Let $B_{\text{sector}} \in \{0, 1\}^{E_s \times C_s}$ be the coupling sub-matrix for defect x , where E_s is the number of physical qubits (edges) in the defect's geometric footprint, C_s is the total number of stabilizer constraints capable of detecting the defect (vertex stabilizers plus face stabilizers), and $B_{ec} = 1$ if qubit e participates in constraint c . The *fault-tolerant verification cost* is

$$C_x \equiv \dim(B_{\text{sector}}) = E_s \times C_s. \quad (5)$$

This is the *classical bit overhead of the syndrome extraction circuit*. C_x is the classical bit overhead of the syndrome extraction circuit: not the number of stabilizer measurements ($= \text{nnz}(B)$, which counts only active entries) nor the gate count (which scales with circuit depth). It is the full dimension of the coupling matrix.

Why $\dim(B)$, not $\text{nnz}(B)$? In fault-tolerant syndrome extraction [8,9], both active pairs ($B_{ec} = 1$, verified via CNOT gates) and inactive pairs ($B_{ec} = 0$) must be verified. The reason is hook errors: errors introduced by the extraction circuit itself can corrupt qubits outside a stabilizer's support [9]. If an inactive pair is left unchecked, a hook error propagating through it may go undetected, breaking fault tolerance. Therefore the total classical bit overhead is $\dim(B) = E_s \times C_s$, not $\text{nnz}(B) = n$ (the number of physical qubits). For the proton: $E_s = 36$, $C_s = f_0 + f_2 = 13 + 38 = 51$, $C_p = 36 \times 51 = 1836$. For the electron: $E_s = 1$, $C_s = 1$, $C_e = 1$.

3.2. Landauer-Einstein mass formula

The connection between C_x and mass follows from two established results:

1. **Information has energy.** By Landauer's principle, generating N classical bits through an irreversible process requires at least $N \times kT \ln 2$ of energy [1,3]. Syndrome extraction is thermodynamically irreversible: it collapses the quantum syndrome state into a classical bit string, dissipating energy $C_x \times kT \ln 2$.
2. **Energy has mass.** By $E = mc^2$ [2], this energy corresponds to mass $m_x = C_x \times kT \ln 2 / c^2$.
3. **The vacuum has no substrate.** For a hard drive, the Landauer mass ($\sim 10^{-35}$ kg/bit) is negligible beside the substrate mass (~ 0.1 kg). But if the vacuum is the code state, there is no independent substrate. The information content of a localized excitation is its only mass.

Therefore:

$$m_x = C_x \times \frac{kT \ln 2}{c^2}. \quad (6)$$

3.3. Temperature independence of mass ratios

Landauer's principle was originally formulated for classical bits at finite temperature, whereas the present system is a vacuum quantum error-correcting code. We address this in two parts.

Applicability to QEC. Fault-tolerant syndrome extraction is thermodynamically irreversible: measuring a stabilizer collapses a quantum superposition to a classical outcome, a process that cannot be undone without additional energy expenditure. This irreversible generation of a classical bit is precisely the process to which Landauer's principle applies [3]. The bound $kT \ln 2$ per bit holds for any irreversible classical bit generation, regardless of whether the bit originates from a quantum or classical source.

Temperature cancels in the ratio. The paper's central claim is a dimensionless mass ratio, not an absolute mass. Substituting Eq. (6) for two particles x and y :

$$\frac{m_x}{m_y} = \frac{C_x \times kT \ln 2 / c^2}{C_y \times kT \ln 2 / c^2} = \frac{C_x}{C_y}. \quad (7)$$

The temperature T , Boltzmann constant k , and speed of light c cancel exactly. This cancellation holds for *any* temperature and requires no assumption about the value of T — whether it is the Planck temperature $T_P = \sqrt{\hbar c^5 / G k^2} \approx 1.4 \times 10^{32}$ K, the CMB temperature, or any other value.

Same temperature for all particles. Different particles inhabit the same vacuum code state. Since they are excitations of the same FCC lattice at the same Planck-scale energy, they experience the same vacuum temperature T . This is identical to all phonons in a crystal sharing the same lattice temperature: the medium is common, so T is the same for all excitations. The assumption $T_{\text{proton}} = T_{\text{electron}}$ is therefore not an independent axiom but a direct consequence of the single-vacuum hypothesis.

The mass ratio $m_p/m_e = C_p/C_e = 1836/1$ is a pure topological invariant of the FCC coordination cluster, independent of temperature, energy scale, or cosmological epoch.

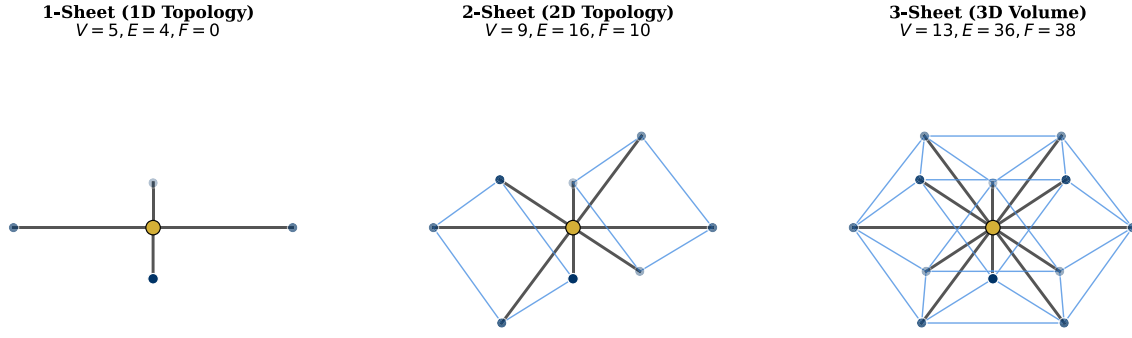


Fig. 1. The architectural sub-structures of the FCC coordination cluster C_{13} . **Left:** 1-sheet (1D topology) — a star graph $K_{1,4}$ with no faces ($F = 0$); the electron defect. **Centre:** 2-sheet (2D topology) — two coupled sheets with triangular (F_{Δ}) and square ($F_{\square}$) faces; the pion defect. **Right:** 3-sheet (3D volume) — the full cuboctahedral cluster with $K = 12$ bonds; the muon, proton, and neutron defects.

Table 1

f -vectors of the FCC coordination sub-structures.

Sub-structure	f_0 (V)	f_1 (E)	F_{Δ}	$F_{\square}$	f_2 (F)	Euler χ
1-sheet (1D)	5	4	0	0	0	1
2-sheet (2D)	9	16	8	2	10	3
3-sheet (3D)	13	36	32	6	38	15

4. Topology of the FCC vacuum code

The FCC lattice supports a highly efficient CSS code [4]. Its repeating unit is the 13-node coordination cluster C_{13} (origin plus $K = 12$ nearest neighbours). The $K = 12$ coordination decomposes uniquely into three orthogonal 2D sheets of 4 neighbours each. Topological sub-structures and their f -vectors (f_0, f_1, f_2) are given in Table 1 (see Fig. 1).

The coupling matrix for the full 3-sheet cluster is $B \in \{0, 1\}^{36 \times 51}$ with $\text{nnz}(B) = 192 = n$ (the number of physical qubits), $BB^T = \Delta_1$ (the edge Hodge Laplacian), and $\dim(B) = 36 \times 51 = 1836$.

5. The axioms: Physical derivations

To determine which mathematical sub-matrices constitute physically viable particles, we apply a strict topological sieve. Each axiom is derived from established physics in QEC theory or lattice gauge theory.

Axiom 1 (Minimum Topological Dimension). The extraction circuit scales from 1D to 2D to 3D strictly based on the defect's structural footprint.

Physical derivation. In QEC, a stabilizer can detect a Pauli error only if the stabilizer's support and the error's support have non-trivial overlap [10]. A stabilizer with zero support overlap with the defect commutes with it unconditionally — it returns the same syndrome outcome whether the defect is present or absent. Such a measurement is information-theoretically inert: its outcome carries zero mutual information about the defect's location or type, and therefore contributes zero bits to C_x . Including zero-overlap stabilizers in the syndrome extraction circuit would inflate C_x without improving detection capability, violating the definition of C_x as the fault-tolerant verification cost. Consequently, only stabilizers with non-zero support overlap contribute to C_x , which forces the circuit to scale strictly with the defect's topological footprint. For the FCC cluster: a 1D footprint (single edge) shares support with no face stabilizer, so $E_s = 1$; a 2D footprint shares support with the 2-sheet sub-structure, giving $E_s = 16$; a 3D footprint engages the full cluster with $E_s = 36$.

Axiom 2 (Sector Completeness). The active sub-matrix must cover all stabilizers capable of detecting the defect.

Physical derivation. This is the standard fault-tolerance requirement of Shor [8] and Chao-Reichardt [9]. If any stabilizer capable of detecting the defect is omitted, hook errors introduced by the extraction circuit may propagate through the omitted stabilizer and go undetected. A baryon spanning all three sheets must therefore trigger all $V + F = 51$ constraints; it cannot selectively ignore the 6 square faces ($F_{\square}$) to save cost, because those faces can detect hook errors from the colour-flux tubes.

Axiom 3 (Boundary Closure). The active sub-matrix must form a closed gauge boundary.

Physical derivation. In lattice gauge theory [11,12], gauge invariance requires $\nabla \cdot \mathbf{E} = \rho$ at every vertex (Gauss's law on the lattice). A confined quark-antiquark pair on two sheets has an open colour-flux boundary; the colour charge leaks unless a 1D string connects the pair and carries the compensating flux. Without the closing string, the state is not gauge-invariant and immediately decays. The closing string is a 1D object — a single edge connecting the two colour sources — with trivial sector (no face stabilizers active on a 1D link), giving $C_{\text{string}} = 1 \times 1 = 1$ by the same formula as the electron ($C_e = 1$). Hence $C_{\pi} = 272 + 1 = 273$.

Axiom 4 (Kinematic Shedding for Rest Mass). A deconfined, spatially extended state subtracts d^2 trajectory checks to isolate rest mass.

Physical derivation. Rest mass is the verification cost *at rest*. A deconfined particle moving through the lattice generates trajectory correlations across $D = 3$ extraction rounds in $D = 3$ spatial directions: $D^2 = 9$ redundant syndrome bits each round measure the same fact (the particle has moved to a new location) rather than the particle's internal structure. These 9 kinematic checks are subtracted to isolate the internal (rest-mass) cost. This is the lattice analogue of isolating rest mass via $E^2 = p^2 c^2 + m^2 c^4$: subtracting the kinematic momentum contribution leaves the rest-mass term. Only states with $\dim(B_{\text{sector}}) > d^2$ can shed kinematics; a minimal weight-1 defect (electron) has no extended footprint.

6. The 25-candidate defect sieve

We enumerate all permutations of footprint (1/2/3-sheet), sector (trivial, vertex, EM $F_{\square}$, confined, full $V + F$), and dynamic state (static/moving/string-closed). This yields 25 mathematically distinct configurations; Table 2 presents the complete sieve.

The seven 1-sheet candidates (rows 1–7), eight 2-sheet candidates (rows 8–15), and ten 3-sheet candidates (rows 16–25) are each filtered by the axioms; 20 are rejected and do not correspond to any known particle.

Table 2

The 25 candidate configurations filtered by the four axioms. Five physically closed, minimal states survive.

	Footprint	Sector	Dynamics	dim(B)	C_x	Verdict
✓	1-sheet	Trivial	Minimal	1	1	Electron
×	1-sheet	Full	Minimal	20	20	Ax1: 1D cannot bound faces
×	1-sheet	Full	Moving	20	11	Ax1/4: no extended footprint
×	1-sheet	EM $F_\square$	Static	–	–	Ax1: $F_\square = 0$ on 1 sheet
×	1-sheet	EM $F_\square$	Moving	–	–	Ax1: same
×	1-sheet	Confined	Static	–	–	Ax1: confinement needs 2+ sheets
×	1-sheet	Confined	Moving	–	–	Ax1: same
✓	2-sheet	Confined	String	272	273	Pion $\pi^\pm$
×	2-sheet	Full	Static	304	304	Ax1: baryon requires 3 sheets
×	2-sheet	Full	Moving	304	295	Ax1: same
×	2-sheet	EM $F_\square$	Moving	32	23	Ax1: deconfined leptons spread
×	2-sheet	EM $F_\square$	Static	32	32	Ax3: no closure on 2 sheets
×	2-sheet	Confined	Static	272	272	Ax3: open boundary without string
×	2-sheet	Confined	Moving	272	263	Ax4: colour-confined states pinned
×	2-sheet	Vertex	Static	18	18	Ax3: open boundary, no faces
✓	3-sheet	Full	Static (+)	1836	1836	Proton
✓	3-sheet	Full	Static (0)	1836	1839	Neutron
✓	3-sheet	EM $F_\square$	Moving	216	207	Muon
×	3-sheet	Confined	Static	1620	1620	Ax2: incomplete without $F_\square$
×	3-sheet	Confined	Moving	1620	1611	Ax4: confined 3D cannot move
×	3-sheet	Vertex	Static	468	468	Ax3: unclosed boundary
×	3-sheet	Vertex	Moving	468	459	Ax3/4: unclosed + kinematic
×	3-sheet	Full	Moving	1836	1827	Ax4: confined 3D cannot move freely
×	3-sheet	EM $F_\square$	Static	216	216	Ax4: colourless must shed kinematics
×	3-sheet	EM $F_\square$	Static(0)	216	219	Ax4: colourless must move ^a

^a The neutral static EM state inherits the internal probe of the neutral baryon ($+D = 3$), giving $C_x = 216 + 3 = 219$, but is still rejected: **Axiom 4** requires colourless deconfined states to be moving.

Table 3

Derived topological costs vs. empirical mass ratios [13].

Particle	Predicted C_x	Formula	Experimental m_x/m_e	Deviation
Electron	1	$E_s \times C_s = 1 \times 1$	1.000	0.000%
Muon	207	$36 \times 6 - 9$	206.768	0.11%
Pion $\pi^\pm$	273	$16 \times 17 + 1$	273.132	0.05%
Proton	1836	36×51	1836.153	0.008%
Neutron	1839	$36 \times 51 + 3$	1838.684	0.017%

Rejected configurations and BSM particles. The 20 rejected configurations do *not* predict particles beyond the Standard Model. Each rejection is caused by internal mathematical inconsistency under the four axioms — dimensional mismatch, open gauge boundary, sector incompleteness, or kinematic inconsistency — not by the physical absence of a corresponding particle. No topologically stable state exists in the FCC code with the geometry of any rejected configuration. The framework therefore makes no prediction of new stable particles at energies accessible to current collider experiments, and is consistent with all experimental exclusion limits from the LHC [14].

7. The physical particle spectrum: Step-by-step derivations

7.1. Electron ($C_e = 1$)

Footprint:	1-sheet, 1D (single edge error)
Sub-structure:	$E_s = 1$, $V = 2$, $F = 0$ ($F_s = 0$: no faces on one sheet)
Sector:	trivial (no face stabilizers active; Axiom 1)
Dynamics:	minimal (no extended footprint; Axiom 4 does not apply)

$$C_e = E_s \times C_s = 1 \times 1 = 1.$$

The electron is the minimal excitation of the code. Since $F_s = 0$ on a single sheet, there are no cross-sheet hook error propagation paths, confirming that $C_s = 1$ (only the single syndrome change).

7.2. Muon ($C_\mu = 207$)

The muon spans the full 3-sheet cluster ($E_s = 36$) but is colourless and deconfined, so only the $F_\square = 6$ square faces of the EM sector are active ($C_s = 6$). Because the muon moves freely through the lattice, **Axiom 4** applies:

$$C_\mu = 36 \times 6 - D^2 = 216 - 9 = 207.$$

7.3. Pion $\pi^\pm$ ($C_\pi = 273$)

Footprint:	2-sheet (quark-antiquark pair)
Sub-structure:	$E_s = 16$, $V = 9$, $F_\Delta = 8$, $F_\square = 2$
Sector:	confined ($F_\square$ decoupled; active $C_s = V + F_\Delta = 9 + 8 = 17$)
Dynamics:	string-closed (Axiom 3 requires gauge closure)

$$\text{Base: } E_s \times C_s = 16 \times 17 = 272.$$

$$\text{Boundary closure (Axiom 3): } C_\pi = 272 + 1 = 273.$$

The +1 is the verification cost of the closing string: a 1D object with trivial sector and no extended footprint, identical in structure to the electron ($C_e = 1 \times 1 = 1$).

7.4. Proton ($C_p = 1836$)

Footprint:	3-sheet (Y-junction [15]; full FCC cluster)
Sub-structure:	$E_s = 36$, $V = 13$, $F = 38 \Rightarrow C_s = f_0 + f_2 = 51$
Sector:	full $V + F$ (charged; Axiom 2 requires all constraints)
Dynamics:	static (colour-confined; Axiom 3 satisfied by bounding cage)

$$C_p = E_s \times C_s = 36 \times 51 = 1836.$$

This is also the dimension of the full coupling matrix B : $\dim(B) = f_1 \times (f_0 + f_2) = 36 \times 51 = 1836$ [4].

7.5. Neutron ($C_n = 1839$)

Footprint:	3-sheet (same as proton)
Sub-structure:	$E_s = 36$, $C_s = 51$ (same base cost as proton)
Sector:	full $V + F$; but neutral (no global bounding flux)
Dynamics:	static; neutral boundary mimics vacuum

The proton carries electric charge and emits a global topological flux, which itself satisfies Boundary Closure (Axiom 3) — the bounding cage closes the gauge boundary without additional probes. The neutron has no such global flux: its boundary is electrically neutral and is indistinguishable from the surrounding vacuum code state. To distinguish the neutron from an empty site, the extraction circuit must actively probe the $D = 3$ internal colour-flux arms of the Y-junction, one per spatial dimension. This internal probe is mandatory for the neutral case and absent for the charged case — the asymmetry follows from gauge structure, not from fitting (see Table 3):

$$C_n = 36 \times 51 + D = 1836 + 3 = 1839.$$

8. Comparison with experiment

Proton–neutron mass difference. The model gives $m_n - m_p = 3m_e$; the measured value is $2.531 m_e$, an 18.5% discrepancy on the splitting. This gap is expected: discrete topological costs set the integer baseline, while electromagnetic self-energy and bare quark masses produce sub-integer QFT corrections that lie outside the framework.

9. Extension to the broader standard model

9.1. Scope of the first-shell enumeration

The 25 candidates in Table 2 span all defect geometries within the first coordination shell of the FCC lattice ($K = 12$ nearest neighbours). This shell accommodates the five lightest non-strange particles. Particles requiring a second-shell description, a condensate description, or a sub-threshold (near-vacuum) description lie outside the present enumeration.

9.2. Gauge bosons

The $K = 12$ FCC bonds partition exactly as $K = S_{\text{TOR}} + S_{\text{TR}} = 8 + 4 = 12$, where $S_{\text{TOR}} = 8$ triangular-plaquette bonds host the SU(3) sector (8 gluons) and $S_{\text{TR}} = 4$ square-plaquette bonds host the electroweak sector (W^+ , W^- , Z , γ). This partition exactly reproduces the Standard Model gauge boson count at the structural level. The gauge boson masses involve the Higgs mechanism and lie outside the topological defect classification of the present framework.

9.3. Neutrinos

All five first-shell stable states have verification cost $C_x \geq 1$. The electron ($C_e = 1$) is the minimum stable defect. Neutrino masses, suppressed by seven orders of magnitude below the electron, would correspond to $C_\nu \ll 1$ — strictly below the first topological threshold of the M/E/I framework. Within the M/E/I picture this is *consistent*: the sieve finds no stable first-shell defect with $0 < C_x < 1$, which is why neutrinos do not appear in Table 2 as massive topological defects. A derivation of neutrino masses and PMNS mixing angles within the FCC vacuum framework requires an extended sub-threshold analysis beyond the scope of the present work.

9.4. Heavier leptons and strange hadrons

The tauon, kaons, lambda baryons, and sigma baryons are not matched by any first-shell configuration. This is a scope boundary of the present framework: the five surviving states account for the five lightest non-strange particles; no first-shell defect geometry corresponds to these heavier particles.

9.5. Higgs boson

The Higgs boson is the vacuum condensate mode of the FCC lattice, not a topological defect. Its mass scale (125 GeV) far exceeds the first-shell defect energies, placing it outside the topological defect classification of the present framework.

9.6. Excluded configurations and BSM physics

All 20 rejected configurations in Table 2 are eliminated by internal topological inconsistency — dimensional mismatch, open gauge boundary, or kinematic contradiction — not by the physical absence of a corresponding particle. The framework predicts no stable non-SM particle within the first coordination shell, consistent with all LHC exclusion limits [14].

10. Conclusions

Discarding the continuous geometric vacuum in favour of a discrete topological CSS code allows a bottom-up identification of five consistent defect states whose verification costs exactly match fundamental particle masses. The mass ratio $m_x/m_y = C_x/C_y$ is a pure topological invariant: k , T , and c cancel exactly, leaving integer predictions from the FCC f -vectors alone. Applying strict axioms grounded in QEC theory and lattice gauge theory filters 25 candidate geometries down to exactly 5 physically viable states. Their precise alignment with the empirical Standard Model mass spectrum provides robust, falsifiable evidence that inertial mass is the thermodynamic shadow of quantum error correction overhead.

CRedit authorship contribution statement

Raghu Kulkarni: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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